Predicting Road Accident Severity (Fatal, Injury, Property Damage Only) on Thailand's Ministry of Transport Road Network, with Cross-Country Validation

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Track	Classification (three-class severity), with cross-country validation
Status	Main project, approved in the instructor consult. Thai data downloaded and profiled (Section 3); cross-country data to be confirmed.

1. Problem Statement

Road accidents cause serious loss of life in Thailand, and predicting how severe an accident is can help prioritise enforcement and emergency response. A 2025 Thai study compared eight classifiers on 112,837 motorcycle accidents (rider at fault, 2015-2020, non-public data from the Department of Disaster Prevention and Mitigation); random forest was best with accuracy 0.777 and AUC 0.768 on a binary fatal vs non-fatal task, but it caught only 19.8% of fatal crashes (recall 0.198) [2]. About 24% of its cases were fatal, so always predicting non-fatal would already score about 76%. We predict a three-class severity (fatal, injury, property damage only) on the Ministry of Transport (MOT) open accident data [1] and report overall and fatal-class metrics. The data of [2] is not public and covers a different population, so we cannot replicate it exactly; we compare in a comparable setting, a motorcycle-only subset of our data. As suggested by the instructor, we also repeat the same pipeline on another country's data to test whether the approach generalises. Other Thai studies use random forest, or k-means with decision trees, on shorter periods [3][4].

2. Research Questions

- **RQ1.** Which factors (time, province, route, vehicle type, cause, accident type, weather) distinguish fatal, injury and property-damage-only accidents?
- **RQ2.** Which model and imbalance strategy (class weights, SMOTE, threshold tuning) gives the best macro-F1 and fatal-class recall, and how does it compare with [2] on a comparable motorcycle-only subset?
- **RQ3.** Does a model trained on 2019-2024 hold up on 2025 and 2026 (temporal drift)?
- **RQ4.** Does the same pipeline beat standard baselines on another country's data (cross-country validation)?

3. Dataset

- **Thai data (confirmed):** Road Accidents on the MOT Road Network (TRAMS), on Open Government Data of Thailand and mirrored on the MOT Data Catalog [1]; Open Data Commons licence, no access restrictions. We downloaded all eight yearly CSV files: **170,022 incident records** from 1 Jan 2019 to 12 Oct 2026 (2026 is a partial year), after removing about 25,000 empty rows. There are 19 columns in 2019 and 38 from 2020: date and time, province, route, first vehicle, road type, presumed cause, accident type, weather, GPS, vehicles and persons involved, deaths and injuries.
- **Target (derived from casualty counts):** fatal (at least one death) 19,437 records (11.4%); injury (no death, at least one injured) 76,431 (45.0%); property damage only 74,154 (43.6%). The fatal share falls from 13.8% in 2019 to 9.9% in 2026. Casualty columns are excluded from the features to avoid leakage.
- **Motorcycle subset (for comparison with [2]):** 26,479 incidents (15.6%) have a motorcycle as the first vehicle. Their fatal share is 30.1% (32.7% in 2019, 28.8% in 2026), close to the 24.4% in [2], so a binary fatal vs non-fatal comparison is meaningful. The MOT data has no rider variables such as gender or drunk, and the first vehicle is not confirmed to be the at-fault one, so this is a comparable setting, not a replication.
- **Data quality issues (handled in preparation):** date formats differ across years (three conventions), 2019 has no serious/minor injury split, 2023 repeats 1,641 accident codes, and the route field is 13% missing.
- **Cross-country data (to be confirmed):** UK STATS19 road-casualty data from the Department for Transport (severity fatal, serious, slight) [5], with prior ML work to compare against [6]; optionally Addis Ababa, Ethiopia police data (12,316 rows; minor, severe, fatal). Labels differ from ours (no property-damage-only class), so the cross-country comparison uses the common fatal vs non-fatal target.

4. Planned Technique (CRISP-DM)

- **Preparation:** merge the yearly files into one schema with per-year date parsing, remove duplicates, handle missing values, encode categorical fields, engineer time and location features; class imbalance handled on training data only.
- **Modelling:** multinomial logistic regression (baseline), decision tree, random forest, XGBoost and LightGBM, plus SVM, k-NN, naive Bayes and a neural network on the motorcycle subset to match the eight algorithms of [2]; hyperparameters tuned with stratified cross-validation inside the training years.

- **Evaluation:** three-class accuracy, macro-F1 and per-class recall against the majority-class baseline (45.0%); PR-AUC and confusion matrices. Temporal split: train 2019-2024, validate 2025, test 2026. For the comparison with [2], a binary fatal vs non-fatal table on the motorcycle subset (also with a random 75:25 split as in [2]) reports accuracy, AUC, and fatal precision and recall beside theirs (random forest AUC 0.768, accuracy 0.777, fatal recall 0.198; k-NN had the best fatal recall, 0.350). Always predicting non-fatal already scores about 76% in [2], 70% on our motorcycle subset and 88.6% on all our data, so we rely on fatal-class recall, F1, AUC and PR-AUC, not accuracy alone.
- **Cross-country:** identical preprocessing, model set and tuning protocol retrained on each country's data (features differ, so no trained weights are transferred); we compare gains over the baselines across countries.
- **Interpretation:** SHAP and feature importance, compared between countries.

5. Expected Contribution

A three-class severity benchmark on 170,022 Thai MOT incidents with temporal validation, a comparison with the 2025 Thai motorcycle study on a comparable subset, a systematic test of imbalance strategies aimed at its weak fatal-crash recall, and evidence from a second country on whether the approach generalises. Reproducible code and a conference-style paper.

6. Timeline

Milestone	Deliverable	Due
#1	Team consult and idea pitch; printed 1-2 page proposal	Week 8 (14-16 Oct 2026)
#2	Dataset confirmed; 1-page proposal submitted	Week 10
#3	Progress check-in: data cleaned, initial results	Week 12
#4	Final presentation, paper (IEEE style), code and dataset	Week 14

References

[1] Road accidents on the Ministry of Transport road network (TRAMS), Open Government Data of Thailand, 2019-2026 (mirror: MOT Data Catalog, datagov.mot.go.th/dataset/roadaccident). https://data.go.th/en/dataset/gdpublish-roadaccident

[2] Mohamad, I., JomnonKwao, S., Ratanavaraha, V. Machine Learning Predictive Performance in Road Accident Severity: A Case Study from Thailand. Results in Engineering, 26, 104833, 2025. DOI 10.1016/j.rineng.2025.104833. https://doi.org/10.1016/j.rineng.2025.104833

[3] Assessing road accident risk levels in Thailand with random forest (accident data 2019-2024). Naresuan University, 2024. https://ww2.agi.nu.ac.th/nred/Document/is-PDF/2567/geo_2567_011_Abstract.pdf

[4] Road Accident Severity Classification in Thailand Using K-Means Clustering and Decision Tree Techniques. https://so03.tci-thaijo.org/index.php/oarit/article/view/296767

[5] UK Department for Transport, road safety data (STATS19), via data.gov.uk; access guide: stats19 R package. https://docs.ropensci.org/stats19/

[6] Geyik, B., Kara, M. Severity Prediction with Machine Learning Methods (STATS19 data), 2020. https://avesis.iku.edu.tr/yayin/85dd12bf-bdab-452c-98db-073a51cc8431/severity-prediction-with-machine-learning-methods

At least 10 references from the last 5 years will be completed for the final paper. The figures quoted from [2] are taken from its full text (Tables 2 and 5); other references were read at abstract level. Cross-country dataset details will be confirmed at download.